

Exercise 6

WM221

Start	1
-1	13

NO.:

DATE:

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State	U	D	L	R	Q
0,0	0	-0.05	0	-0.05	
0,1	0	0.5	0	0.5	
0,2	0	0.7	0	0	
1,0	0	0	0	-0.05	
1,1	0	0	0	-0.05	
1,2	0	2.2	0	0	
2,0	0	0	0	0	
2,1	0	0	0	0	
2,2	0	0	0	0	

Sarsa 1st ep:

Step 1: ~~$Q(0,0,D) = 0 + 0.5(-0.1 - 0) = -0.05$~~

target = $-0.1 + Q(1,0,D) = -0.1 + 0 = -0.1$

$Q(0,0,D) = 0 + 0.5(-0.1 - 0) = -0.05$

Step 2:

target = $-0.1 + Q(1,1,R) = 0.1 + 0 = 0.1$

$Q(1,0,R) = 0 + 0.5(-0.1 - 0) = -0.05$

Step 3:

target = $-0.1 + 0 = -0.1$

$Q(1,1,R) = 0 + 0.5(-0.1 - 0) = -0.05$

Step 4:

~~$Q(1,2,D) = 0 + 0.5(3 - 0) = 1.5$~~

$Q(1,2,D) = 0 + 0.5(3 - 0) = 1.5$

Sarsa 2nd Ep:

Step 1:

target = $-0.1 + Q(0,1,R) = -0.1 + 0 = -0.1$

$Q(0,0,R) = 0 + 0.5(-0.1 - 0) = -0.05$

Step 2:

target = $1 + Q(0,2,R) = 1 + 0 = 1$

$Q(0,1,R) = 0 + 0.5(1 - 0) = 0.5$

Step 3:

target = ~~$-0.1 + 1.5 = 1.4$~~

$Q(0,2,R) = 0 + 0.5(1.4 - 0) = 0.7$

Step 4:

target =

$Q(1,2,D) = 1.5 + 0.5(3 - 1.5) = 2.25$

Q Learning

State	U	D	L	R	Q-Value 1st ep:
0,0	0	-0.05	0	-0.05	target 1st step:
0,1	0	0	0	0.5	target = $-0.1 + Q(0,1,0) = -0.1 + 0 = -0.1$
0,2	0	0.7	0	0	$Q(0,0,D) = 0 + 0.5(-0.1 - 0) = -0.05$
1,0	0	0	0	-0.05	Step 2:
1,1	0	0	0	-0.05	target = $-0.1 + Q(1,1,R) = -0.1 + 0 = -0.1$
1,2	0	2.25	0	0	$Q(1,0,R) = 0 + 0.5(-0.1 - 0) = -0.05$
2,0	0	0	0	0	Step 3:
2,1	0	0	0	0	target = $-0.1 + \text{0} = \text{-0.1}$
2,2	0	0	0	0	$Q(1,1,R) = 0 + 0.5(-0.1 - 0) = -0.05$

Step 4

$$Q(1,2,D) = 0 + 0.5(3 - 0) = 1.5$$

Episode 2:

1st step: $-0.1 + Q(0,1,R) = -0.1 + 0 = -0.1$

Target = $0 + 0.5(-0.1 - 0 - 0) = \text{~~-0.05~~} - 0.05$

$Q(0,0,R) = \text{~~0 + 0.5(-0.1 - 0)~~}$

Step 2:

Target = $1 + Q(0,2,R) = 1 + 0 = 1$

$Q(0,1,R) = \text{~~0 + 0.5(1 - 0)~~} = 0 + 0.5(1 - 0) = 0.5$

Step 3:

Target = $-0.1 + 1.5 = 1.4$

$Q(0,2,D) = 0 + 0.5(1.4 + 0) = 0.7$

Step 4

$Q(1,2,D) = 1.5 + 0.5(3 - 1.5) = 2.25$